

Photons

23

In this chapter, we will consider the thermodynamics of electromagnetic radiation. It was Maxwell who realized that light was an electromagnetic wave and that the speed of light, c , could be expressed in terms of fundamental constants taken from the theories of electricity and magnetism. In modern notation, this relation is

$$c = 1/\sqrt{\epsilon_0\mu_0}, \quad (23.1)$$

where ϵ_0 and μ_0 are the permittivity and permeability of free space respectively. Later, Planck realized that light behaved not only like a *wave* but also like a *particle*. In the language of quantum mechanics, electromagnetic waves can be *quantized* as a set of particles which are known as **photons**. Each photon has an energy $\hbar\omega$ where $\omega = 2\pi\nu$ is the angular frequency.¹ Each photon has a momentum $\hbar k$ where k is the wave vector.² The ratio of the energy to the momentum of a photon is

$$\frac{\omega}{k} = 2\pi\nu \times \frac{\lambda}{2\pi} = \nu\lambda = c. \quad (23.2)$$

Electromagnetic radiation is emitted from any substance at non-zero temperature. This is known as **thermal radiation**. For objects at room temperature, you may not have noticed this effect because the frequency of the electromagnetic radiation is low and most of the emission is in the infrared region of the electromagnetic spectrum. Our eyes are only sensitive to electromagnetic radiation in the visible region. However, you may have noticed that a piece of metal in a furnace glows ‘red hot’ so that, for such objects at higher temperature, your eyes are able to pick up some of the thermal radiation.³

This chapter is all about the properties of this thermal radiation. We will begin in Sections 23.1–23.4 by restricting ourselves to simple thermodynamics arguments to derive as much as we can about thermal radiation without going into the gory details, in much the same way as was originally done in the nineteenth century. This approach doesn’t get us the whole way, but provides a lot of insight. Then in Sections 23.5–23.6, we will use the more advanced statistical mechanical techniques introduced in the previous chapters to do the job properly. The final sections concern the thermal radiation that exists in the Universe as a remnant of the hot Big Bang and the effect of thermal radiation on the behaviour of atoms and hence the operation of the laser.

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¹ ν is the frequency. The energy can also be expressed as $\hbar\nu$. Recall also that $\hbar = h/(2\pi)$.

² The wave vector $k = 2\pi/\lambda$ where λ is the wavelength.

³ Your eyes can pick up a lot of the thermal radiation if they are assisted by infrared goggles.

23.1 The classical thermodynamics of electromagnetic radiation

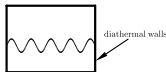


Fig. 23.1 A cavity of photons whose walls are diathermal, meaning they are in thermal contact with their surroundings, so that the temperature within may be controlled.

In this section, we will consider the thermodynamics of electromagnetic radiation from a classical standpoint, although we will allow ourselves the post-nineteenth century luxury of considering the electromagnetic radiation to consist of a gas of photons. First we will consider the effect of a collection of photons on the surroundings which contain it. Let us consider the surroundings to be a container of volume V , which in this subject is termed a 'cavity', which is held at temperature T . The photons inside the cavity are in thermal equilibrium with the cavity walls, and form electromagnetic standing waves. The walls of the cavity, shown in Fig. 23.1, are made of diathermal material (i.e. they transmit heat between the gas of photons inside the cavity and the surroundings). If n photons per unit volume comprise the gas of photons in the cavity then the energy density u of the gas may be written as:

$$u = \frac{U}{V} = n\hbar\omega, \quad (23.3)$$

where $\hbar\omega$ is the mean energy of a photon. From kinetic theory (eqn 6.15), the pressure p of a gas of particles is $\frac{1}{3}nm\langle v^2 \rangle$. For photons, we replace $\langle v^2 \rangle$ in this formula by c^2 , the square of the speed of light. Interpreting mc^2 as the energy of a photon, we then have that p is one third of the energy density. Thus

$$p = \frac{u}{3}, \quad (23.4)$$

which is different from the expression in eqn 6.25 ($p = 2u/3$) from the kinetic theory of gases, a point which we will return to in Section 25.2 (see eqn 25.21).⁴ Equation 23.4 gives an expression for the radiation pressure due to the electromagnetic radiation. Also from kinetic theory (eqn 7.6), the flux Φ of photons on the walls of their container, that is to say the number of photons striking unit area of their container per second, is given by

$$\Phi = \frac{1}{4}nc, \quad (23.5)$$

where c is the speed of light. From this, and eqn 23.3, we can write the power incident per unit area of cavity wall, due to the photons, as

$$P = \hbar\omega\Phi = \frac{1}{4}uc. \quad (23.6)$$

This relation will be important as we now derive the Stefan-Boltzmann law, which relates the temperature of a body to the energy flux radiating from it in the form of electromagnetic radiation. We can derive this using the first law of thermodynamics in the form $dU = TdS - pdV$ to give

$$\begin{aligned} \left(\frac{\partial U}{\partial V} \right)_T &= T \left(\frac{\partial S}{\partial V} \right)_T - p \\ &= T \left(\frac{\partial p}{\partial T} \right)_V - p, \end{aligned} \quad (23.7)$$

⁴The factor of two difference arises from writing the kinetic energy as mc^2 and not as $\frac{1}{2}m\langle v^2 \rangle$, and thus reflects the difference in form between the equation for the relativistic energy of a photon and that for the kinetic energy of a non-relativistic particle.

where the last equality follows from using a Maxwell relation. The left-hand side of eqn 23.7 is simply⁵ the energy density u . Hence, using eqn 23.7, together with eqn 23.4, we obtain

$$u = \frac{1}{3}T \left(\frac{\partial u}{\partial T} \right)_V - \frac{u}{3}. \quad (23.8)$$

Rearranging gives:

$$4u = T \left(\frac{\partial u}{\partial T} \right)_V, \quad (23.9)$$

from which follows

$$4 \frac{dT}{T} = \frac{du}{u}. \quad (23.10)$$

Equation 23.10 may be integrated to give:

$$u = AT^4, \quad (23.11)$$

where A is a constant of the integration with units $\text{JK}^{-1} \text{m}^{-3}$. We can now use eqn 23.6 to give us the power incident⁶ per unit area.⁷

$$P = \frac{1}{4}uc = \left(\frac{1}{4}Ac \right) T^4 = \sigma T^4, \quad (23.12)$$

where the term in brackets, $\sigma = \frac{1}{4}Ac$, is known as the **Stefan–Boltzmann constant**. Equation 23.12 is known as the **Stefan–Boltzmann law** or sometimes as **Stefan's law**. For the moment, we have no idea what value the constant σ takes and this is something that was originally determined from experiment. In Section 23.5, using the techniques of statistical mechanics, we will derive an expression for this constant.

23.2 Spectral energy density

The energy density u of electromagnetic radiation is a quantity which tells you how many Joules are stored in a cubic metre of cavity. What we want to do now is to specify in which frequency ranges that energy is stored. All of this will fall out of the statistical mechanical treatment in Section 23.5, but we want to continue to apply a classical treatment to see how far we can get. To do this, consider two containers, each in contact with thermal reservoirs at temperature T and joined to one another by a tube, as illustrated schematically in Fig. 23.2. The system is allowed to come to equilibrium.

The thermal reservoirs are at the same temperature T and so we know from the second law of thermodynamics that there can be no net heat flow from either one of the bodies to the other. Therefore there can be no net energy flux along the tube, so that the energy flux from the soot-lined cavity along the tube from left to right must be balanced by the energy flux from the mirror-lined cavity along the tube from right to left. Equation 23.12 thus tells us that each cavity must have the same energy density u . This argument can be repeated for cavities of different

⁵This should be obvious since it is the definition of energy density. However, if you want to convince yourself, notice that differentiating $U = uV$ with respect to V yields

$$\left(\frac{\partial U}{\partial V} \right)_T = u + V \left(\frac{\partial u}{\partial V} \right)_T = u,$$

because $\left(\frac{\partial u}{\partial V} \right)_T = 0$ since u , an energy density, is independent of volume.

⁶Note that when the cavity is in equilibrium with the radiation inside it, the power incident is equal to the power emitted; hence the expression for P expresses the power emitted by the surface and the power incident on the surface.

⁷The power per unit area is equal to an energy flux.

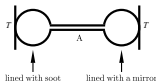


Fig. 23.2 Two cavities at temperature T , one is lined with soot and the other with a mirror coating.

shape and size as well as different coatings. Hence we conclude that u is independent of shape, size or material of the cavity. But maybe one cavity might have more energy density than the other at certain wavelengths, even if it has to have the same energy density overall? This is not the case, as we shall now prove. First, we make a definition.

u_λ has units $\text{J m}^{-3} \text{m}^{-1}$.

We can also define a spectral density in terms of frequency ν , so that $u_\nu d\nu$ is the energy density due to those photons which have frequencies between ν and $\nu + d\nu$.

- The **spectral energy density** u_λ is defined as follows: $u_\lambda d\lambda$ is the energy density due to those photons which have wavelengths between λ and $\lambda + d\lambda$. The total energy density is then

$$u = \int u_\lambda d\lambda. \quad (23.13)$$

Now imagine that a filter, which only allows a narrow band of radiation at wavelength λ to pass, is inserted at point A in Fig. 23.2 and the system is left to come to equilibrium. The same arguments listed above apply in this case: there is no net energy flux from one cavity to the other and hence the specific internal energy within a narrow wavelength range is the same for each case:

$$u_\lambda^{\text{soot}}(T) = u_\lambda^{\text{mirror}}(T). \quad (23.14)$$

This demonstrates that the spectral internal energy has no dependence on the material, shape, size or nature of a cavity. The spectral energy density is thus a universal function of λ and T only.

23.3 Kirchhoff's law

We now wish to discuss how well particular surfaces of a cavity will absorb or emit electromagnetic radiation of a particular frequency or wavelength. We therefore make the following additional definitions:

α_λ is dimensionless.

e_λ has units $\text{W m}^{-2} \text{m}^{-1}$.

- The **spectral absorptivity** α_λ is the fraction of the incident radiation which is absorbed at wavelength λ .
- The **spectral emissive power** e_λ of a surface is a function such that $e_\lambda d\lambda$ is the power emitted per unit area by the electromagnetic radiation having wavelengths between λ and $\lambda + d\lambda$.

Using these definitions, we may now write down the form for the power per unit area absorbed by a surface, if the incident spectral energy density is $u_\lambda d\lambda$, as follows:

$$\left(\frac{1}{4} u_\lambda d\lambda c \right) \alpha_\lambda. \quad (23.15)$$

The power per unit area emitted by a surface is given by

$$e_\lambda d\lambda. \quad (23.16)$$

In equilibrium, the expressions in eqns 23.15 and 23.16 must be equal, and hence

$$\frac{e_\lambda}{\alpha_\lambda} = \frac{c}{4} u_\lambda. \quad (23.17)$$

Equation 23.17 expresses **Kirchhoff's law**, which states that the ratio e_λ/α_λ is a universal function of λ and T . Therefore, if you fix λ and T , the ratio e_λ/α_λ is fixed and hence $e_\lambda \propto \alpha_\lambda$. In other words 'good absorbers are good emitters' and 'bad absorbers are bad emitters'.

Example 23.1

Dark coloured objects which absorb most of the light that falls on them will be good at emitting thermal radiation. One has to be a bit careful here because you have to be sure about which wavelength you are talking about. A better statement of Kirchhoff's laws would be 'good absorbers at one wavelength are good emitters at the same wavelength'.

For example, a white coffee mug absorbs poorly in visible wavelengths so looks white. A black, but otherwise identical, coffee mug absorbs well in visible wavelengths so looks black. Which one is best at keeping your coffee warm? You might conclude that it is the white mug because 'poor absorbers are poor emitters' and that the mug will lose less heat by thermal radiation. However, a hot mug emits radiation mainly in the infrared region of the electromagnetic spectrum,⁸ and so the mug being white in the visible is immaterial; what you need to know is what 'colour' each mug is in the infrared, i.e. measuring their absorption spectra at infrared wavelengths will tell you about their emission properties there.

⁸See Appendix D.

A perfect **black body** is an object which is defined to have $\alpha_\lambda = 1$ for all λ . Kirchhoff's law expressed in eqn 23.17 tells us that for this maximum value of α , a black body is the best possible emitter. It is often useful to think of a **black body cavity** which is an enclosure whose walls have $\alpha_\lambda = 1$ for all λ and which contains a gas of photons at the same temperature as the walls, due to emission and absorption of photons by the atoms in the walls. The gas of photons contained in the black body cavity is known as **black body radiation**.

Example 23.2

The temperature of the Earth's surface is maintained by radiation from the Sun. By making the approximation that the Sun and the Earth behave as black bodies, show that the ratio of the Earth's temperature to that of the Sun is given by

$$\frac{T_{\text{Earth}}}{T_{\text{Sun}}} = \sqrt{\frac{R_{\text{Sun}}}{2D}}, \quad (23.18)$$

where R_{Sun} is the radius of the Sun and the Earth-Sun separation is D .
Solution:

The Sun emits a power equal to its surface area $4\pi R_{\text{Sun}}^2$ multiplied by σT_{Sun}^4 . This power is known as its **luminosity** L (measured in Watts), so that

$$L = 4\pi R_{\text{Sun}}^2 \sigma T_{\text{Sun}}^4. \quad (23.19)$$

At a distance D from the Sun, this power is uniformly distributed over a sphere with surface area $4\pi D^2$, and the Earth is only able to ‘catch’ this power over its projected area πR_{Earth}^2 . Thus the power incident on the Earth is

$$\text{power incident} = L \left(\frac{\pi R_{\text{Earth}}^2}{4\pi D^2} \right). \quad (23.20)$$

The power emitted by the Earth, assuming it has a uniform temperature T_{Earth} and behaves as a black body, is simply $\sigma T_{\text{Earth}}^4$ multiplied by the Earth’s surface area $4\pi R_{\text{Earth}}^2$, so that

$$\text{power emitted} = 4\pi R_{\text{Earth}}^2 \sigma T_{\text{Earth}}^4 \quad (23.21)$$

Equating eqn 23.20 and eqn 23.21 yields the desired result.

Putting in the numbers $R_{\text{Sun}} = 7 \times 10^8$ m, $D = 1.5 \times 10^{11}$ m and $T_{\text{Sun}} = 5800$ K yields $T_{\text{Earth}} = 280$ K, which is not bad given the crudeness of the assumptions.

23.4 Radiation pressure

To summarize the results of the earlier sections in this chapter, for black body radiation we have:

$$\text{power radiated per unit area} \quad P = \frac{1}{4}uc = \sigma T^4, \quad (23.22)$$

$$\text{energy density in radiation} \quad u = \left(\frac{4\sigma}{c} \right) T^4, \quad (23.23)$$

$$\text{pressure on cavity walls} \quad p = \frac{u}{3} = \frac{4\sigma T^4}{3c}. \quad (23.24)$$

If, however, one is dealing with a beam of light, in which all the photons are going in the same direction (rather than in each and every direction as we have in a gas of photons) then these results need to be modified. The pressure exerted by a collimated beam of light can be calculated as follows: a cubic metre of this beam has momentum $nhk = n\hbar\omega/c$, and this momentum is absorbed by a unit area of surface, normal to the beam, in a time $1/c$. Thus the pressure is $p = [n\hbar\omega/c]/[1/c] = n\hbar\omega = u$. A cubic metre of the beam has energy $n\hbar\omega$, so the power P incident on unit area of surface is $P = n\hbar\omega/(1/c) = uc$. Hence, we have

$$\text{power radiated per unit area} \quad P = uc = \sigma T^4, \quad (23.25)$$

$$\text{energy density in radiation} \quad u = \left(\frac{\sigma}{c} \right) T^4, \quad (23.26)$$

$$\text{pressure on cavity walls} \quad p = u = \frac{\sigma T^4}{c}. \quad (23.27)$$

It is worth emphasising that electromagnetic radiation exerts a real pressure on a surface and this can be calculated using eqn 23.24 or eqn 23.27 as appropriate. An example of a calculation of radiation pressure is given below.

Example 23.3

Sunlight falls on the surface of the Earth with a power per unit area equal to $P = 1370 \text{ W m}^{-2}$. Calculate the radiation pressure and compare it to atmospheric pressure.

Solution:

Sunlight on the Earth's surface consists of photons all going in the same direction,⁹ and hence we can use

$$p = \frac{P}{c} = 4.6 \mu\text{Pa}, \quad (23.28)$$

which is more than ten orders of magnitude lower than atmospheric pressure (which is $\sim 10^5 \text{ Pa}$).

⁹We make this approximation because the Sun is sufficiently far from the Earth, so that all the rays of light arriving on Earth are parallel.

23.5 The statistical mechanics of the photon gas

Our argument so far has only used classical thermodynamics. We have been able to predict that the energy density u of a photon gas behaves as AT^4 but we have been able to say nothing about the constant A . It was only through the development of quantum theory that it was possible to derive what A is, and we will present this in what follows. The crucial insight is that electromagnetic waves in a cavity can be described by simple harmonic oscillators. The angular frequency ω of each mode of oscillation is related to the wave vector k by

$$\omega = ck \quad (23.29)$$

(see Fig. 23.3) and hence the density of states¹⁰ of electromagnetic waves as a function of wave vector k is given by

$$g(k) dk = \frac{4\pi k^2 dk}{(2\pi/L)^3} \times 2, \quad (23.30)$$

where the cavity is assumed to be a cube of volume $V = L^3$ and the factor 2 corresponds to the two possible polarizations of the electromagnetic waves. Thus

$$g(k) dk = \frac{V k^2 dk}{\pi^2}, \quad (23.31)$$

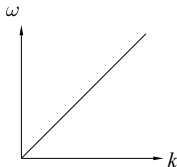


Fig. 23.3 The relation between ω and k , for example that in eqn 23.29, is known as a dispersion relation. For light (plotted here) this relation is very simple and is called non-dispersive because both the phase velocity (ω/k) and the group velocity ($d\omega/dk$) are equal.

¹⁰This treatment is similar to the analysis in Section 21.1 for the ideal gas.

and hence the density of states $g(\omega)$, now written as a function of frequency using eqn 23.29, is

$$g(\omega) = g(k) \frac{dk}{d\omega} = \frac{g(k)}{c}, \quad (23.32)$$

and hence

$$g(\omega) d\omega = \frac{V\omega^2 d\omega}{\pi^2 c^3}. \quad (23.33)$$

We can derive U for the photon gas by using the expression for U for a single simple harmonic oscillator in eqn 20.29 to give

$$U = \int_0^\infty g(\omega) d\omega \hbar\omega \left(\frac{1}{2} + \frac{1}{e^{\beta\hbar\omega} - 1} \right). \quad (23.34)$$

This presents us with a problem since the first part of this expression, due to the sum of all the zero-point energies, diverges:

$$\int_0^\infty g(\omega) d\omega \frac{1}{2} \hbar\omega \rightarrow \infty. \quad (23.35)$$

This must correspond to the energy of the vacuum, so after swallowing hard we redefine our zero of energy so that this infinite contribution is swept conveniently under the carpet. We are therefore left with

$$U = \int_0^\infty g(\omega) d\omega \frac{\hbar\omega}{e^{\beta\hbar\omega} - 1} = \frac{Vh}{\pi^2 c^3} \int_0^\infty \frac{\omega^3 d\omega}{e^{\beta\hbar\omega} - 1}. \quad (23.36)$$

If we make the substitution $x = \hbar\beta\omega$, we can rewrite this as

$$U = \frac{Vh}{\pi^2 c^3} \left(\frac{1}{\hbar\beta} \right)^4 \int_0^\infty \frac{x^3 dx}{e^x - 1} = \left(\frac{V\pi^2 k_B^4}{15c^3 \hbar^3} \right) T^4, \quad (23.37)$$

and hence $u = U/V = AT^4$. Here, use has been made of the integral

$$\int_0^\infty \frac{x^3 dx}{e^x - 1} = \zeta(4)\Gamma(4) = \frac{\pi^4}{15}, \quad (23.38)$$

which is proved in Appendix C.4 (see eqn C.25). This therefore establishes that the constant $A = 4\sigma/c$ is given by

$$A = \frac{\pi^2 k_B^4}{15c^3 \hbar^3}, \quad (23.39)$$

¹¹If you prefer to use h , rather than $\hbar$, the **Stefan-Boltzmann constant** is written as

$$\sigma = \frac{2\pi^5 k_B^4}{15c^2 h^3}.$$

and hence the Stefan-Boltzmann constant¹¹ σ is

$$\sigma = \frac{\pi^2 k_B^4}{60c^2 \hbar^3} = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}. \quad (23.40)$$

23.6 Black body distribution

The expression in eqn 23.36 can be rewritten as

$$u = \frac{U}{V} = \int u_\omega d\omega, \quad (23.41)$$

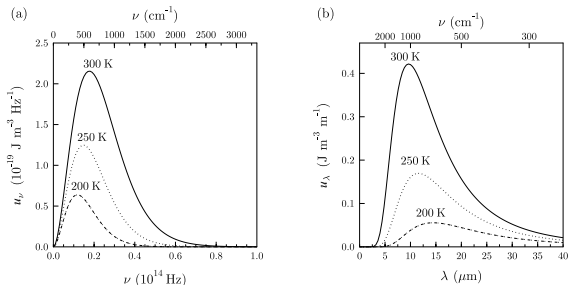


Fig. 23.4 The black body distribution of spectral energy density, plotted for 200 K, 250 K and 300 K as a function of (a) frequency and (b) wavelength. The upper scale shows the frequency in inverse centimetres, a unit beloved of spectroscopists.

where u_ω is a different form of the spectral energy density (written this time as a function of angular frequency $\omega = 2\pi\nu$). It thus takes the form

$$u_\omega = \frac{\hbar}{\pi^2 c^3} \frac{\omega^3}{e^{\beta\hbar\omega} - 1}. \quad (23.42)$$

This spectral energy density function is known as a **black body distribution**. We can also express this in terms of frequency ν by writing $u_\omega d\omega = u_\nu d\nu$, and using $\omega = 2\pi\nu$ and hence $d\omega/d\nu = 2\pi$. This yields

$$u_\nu = \frac{8\pi h}{c^3} \frac{\nu^3}{e^{\beta h\nu} - 1}. \quad (23.43)$$

This function is plotted in Fig. 23.4(a). Similarly, we can transform this into wavelength, by writing $u_\nu d\nu = u_\lambda d\lambda$, and using $\nu = c/\lambda$ and hence $d\nu/d\lambda = -c/\lambda^2$. This yields an expression for u_λ as follows:

$$u_\lambda = \frac{8\pi hc}{\lambda^5} \frac{1}{e^{\beta hc/\lambda} - 1}. \quad (23.44)$$

This is shown in Fig. 23.4(b).

We note several features of this black body distribution.

- At low frequency (i.e. long wavelength), when $h\nu/k_B T \ll 1$, the exponential term can be written as

$$e^{\beta h\nu} \approx 1 + \frac{h\nu}{k_B T}, \quad (23.45)$$

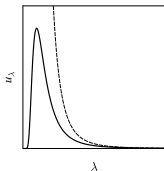


Fig. 23.5 The black body energy density u_λ (thick solid line), together with the Rayleigh-Jeans equation (eqn 23.47) which is the long-wavelength limit of the black body distribution.

and hence

$$u_\nu \rightarrow \frac{8\pi k_B T \nu^2}{c^3}, \quad (23.46)$$

and equivalently

$$u_\lambda \rightarrow \frac{8\pi k_B T}{\lambda^4}. \quad (23.47)$$

These two expressions are different forms of the **Rayleigh-Jeans law**, and were derived in the nineteenth century before the advent of quantum mechanics. As that might imply, Planck's constant h does not appear in them. These expressions are the correct limit of the black body distribution, as shown in Fig. 23.5. They created problems at the time, because if you take the Rayleigh-Jeans form of u_λ and assume it is true for all wavelengths, and then try and integrate it to get the total internal energy U , you find that

$$U = \int_0^\infty u_\lambda d\lambda = \int_0^\infty \frac{8\pi k_B T d\lambda}{\lambda^4} \rightarrow \infty. \quad (23.48)$$

This apparent divergence in U was called the **ultraviolet catastrophe**, because integrating down to small wavelengths (towards the ultraviolet) produced a divergence. In fact, such high-energy electromagnetic waves are not excited because light is quantized and it costs too much energy to produce an ultraviolet photon when the temperature is too low. Of course, using the correct black body u_λ from eqn 23.44, the correct form

$$U = \int_0^\infty u_\lambda d\lambda = \frac{4\sigma}{c} T^4 \quad (23.49)$$

is obtained.

- One can also define the **radiance** (or **surface brightness**) B_ν as the flux of radiation per steradian (the unit of solid angle, abbreviated to sr) in a unit frequency interval. This function gives the power *through* an element of unit area, per unit frequency, *from* an element of solid angle. The units of radiance are $\text{W m}^{-2} \text{Hz}^{-1} \text{sr}^{-1}$. Because there are a total of 4π steradians, we have that¹²

$$B_\nu(T) = \frac{c}{4\pi} u_\nu(T) = \frac{2h}{c^2} \frac{\nu^3}{e^{\beta h \nu} - 1}. \quad (23.50)$$

By analogy, B_λ , with units $\text{W m}^{-2} \text{m}^{-1} \text{sr}^{-1}$, is defined by

$$B_\lambda(T) = \frac{c}{4\pi} u_\lambda(T) = \frac{2hc^2}{\lambda^5} \frac{1}{e^{\beta h c / \lambda} - 1}. \quad (23.51)$$

- Wien found experimentally in 1896, before the advent of quantum mechanics, that the product of the temperature and of the wavelength at which the maximum of the black body distribution u_λ is found is a constant. This is a statement of what is known as **Wien's law**. The constant can be given as follows:

$$\lambda_{\text{max}} T = \text{a constant}. \quad (23.52)$$

¹²Note that if we divide the energy density by the time taken for unit volume of photons to pass through unit area of surface, namely $1/c$, we have the energy flux.

Wien's law follows from the fact that $\lambda_{\max}$ can be determined by the condition $du_{\lambda}/d\lambda = 0$, and applying this to eqn 23.44 leads to $\beta hc/\lambda_{\max} = \text{a constant}$. Hence $\lambda_{\max}T$ is a constant, which is Wien's law. The law tells us that at room temperature, objects which are approximately black bodies will radiate the most at wavelength $\lambda_{\max} \approx 10\mu\text{m}$, which is in the infrared region of the electromagnetic spectrum, as demonstrated in Fig. 23.4(b). One can easily show¹³ that the maximum in u_{ν} occurs at a frequency given by

$$\frac{h\nu}{k_{\text{B}}T} = 2.82144 \quad (23.53)$$

and the maximum in u_{λ} occurs at a wavelength given by

$$\frac{hc}{\lambda k_{\text{B}}T} = 4.96511. \quad (23.54)$$

This can be used to show that the product λT is given by

$$\lambda T = \begin{cases} 5.1 \text{ mm K} & \text{at the maximum of } u_{\nu}(T), \\ 2.9 \text{ mm K} & \text{at the maximum of } u_{\lambda}(T). \end{cases} \quad (23.55)$$

These maxima do not occur at the same place for each distribution because one is measured per unit frequency interval and the other per unit wavelength interval, and these are different.¹⁴

Figure 23.6(a) shows how the shape of the distribution changes with temperature for u_{ν} and Fig. 23.6(b) for u_{λ} on log-log scales. These diagrams show how the peak of the black body distribution lies in the optical region of the spectrum for temperatures of several thousand Kelvin, but in the microwave region for a temperature of a few Kelvin. This fact is very relevant for the black body radiation in the Universe, which we describe in the following section.

23.7 Cosmic Microwave Background radiation

In 1978, Penzias and Wilson of Bell Labs, New Jersey, USA won the Nobel Prize for their serendipitous discovery (in 1963–1965) of seemingly uniform microwave emission coming from all directions in the sky which has come to be known as the **cosmic microwave background (CMB)**. Remarkably, the spectral shape of this emission exhibits, to high precision, the distribution for black body radiation of temperature 2.7 K (see Fig. 23.7) with a peak in the emission spectrum at a wavelength of about 1 mm. It is startling that the radiation is uniform, or isotropic, to better than 1 part in 10^5 (meaning that its spectrum and intensity is almost the same if you measure in different directions in the sky). This is one of the key pieces of evidence in favour of the hot Big Bang model for the origin of the Universe. It implies that there was a time when all of the Universe we see now was in thermal equilibrium.¹⁵

¹³See Exercise 23.2.

¹⁴The difference between $d\nu$ and $d\lambda$ is derived as follows:

$$c = \nu\lambda,$$

and hence

$$\nu = c/\lambda,$$

so that

$$d\nu = -\frac{c}{\lambda^2}d\lambda.$$

¹⁵Note that different black body distributions, that is multiple curves corresponding to regions at a variety of different temperatures, do not superpose to form a single black body distribution.

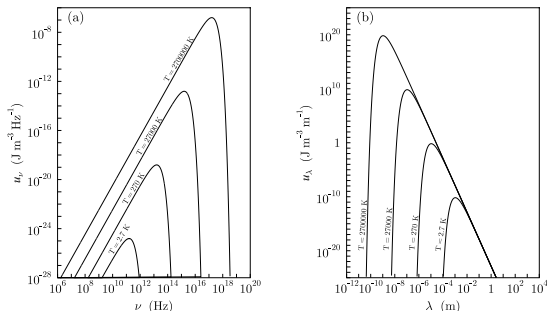


Fig. 23.6 The black body distribution of spectral energy density, plotted on a logarithmic scale for four different temperatures as a function of (a) frequency and (b) wavelength.

We can make various inferences about the origin of the Universe from observations of the cosmic microwave background. It can be shown that the energy density of radiation in the expanding Universe falls off as the fourth power of the scale factor (which you can think of as the linear magnification factor describing the separation of a pair of marker galaxies in the Universe, a quantity which increases with cosmic time). From the Stefan-Boltzmann law, the energy density of radiation falls off as T^4 , so temperature and scale-factor are inversely proportional to one another, so the Universe cools as it expands. Conversely, when the Universe was much younger, it was much smaller and much hotter. Extrapolating back in time, one finds that temperatures were such that physical conditions were very different. For example, it was too hot for matter to exist as atoms, and everything was ionized. Further back in cosmic time still, even quarks and hadrons, the sub-structure of protons and neutrons were thought to be dissociated.

23.8 The Einstein A and B coefficients

If a gas of atoms is subjected to thermal radiation, the atoms can respond by making transitions between different energy levels. We can think about this effect in terms of absorption and emission of photons

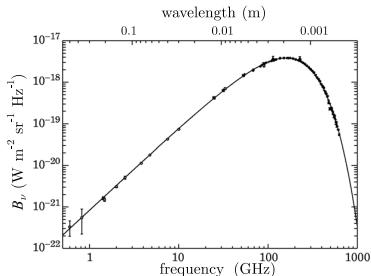


Fig. 23.7 The experimentally determined spectrum of the cosmic microwave background (data courtesy NASA).

by the atom. The atoms are sitting in a bath of photons which we call the **radiation field** and it has an energy density u_ω given by eqn 23.42. In this section, we will consider the effect of this radiation field on the transitions between atomic energy levels by modelling the atom as a simple two-level system. Consider the two-level system shown in Fig. 23.8 which comprises two energy levels, a lower level 1 and an upper level 2, separated by an energy $\hbar\omega$. In the absence of the radiation field, atoms in the upper level can decay to the lower level by the process of **spontaneous emission** of a photon [Fig. 23.8(a)]. The number of atoms in the upper level, N_2 , is given by solving a simple differential equation

$$\frac{dN_2}{dt} = -A_{21}N_2, \quad (23.56)$$

where A_{21} is a constant. This expresses simply that the decay rate depends on the number of atoms in the upper level. The solution of this equation is

$$N_2(t) = N_2(0)e^{-t/\tau}, \quad (23.57)$$

where $\tau \equiv 1/A_{21}$ is the **natural radiative lifetime** of the upper level.

In the presence of a radiation field of energy density u_ω , two further processes are possible:

- An atom in level 1 can absorb a photon of energy $\hbar\omega$ and will end up in level 2 [Fig. 23.8(b)]. This process is called **absorption**, and will occur at a rate which is proportional both to u_ω and to the number of atoms in level 1. Thus the rate can be written as $N_1 B_{12} u_\omega$, where B_{12} is a constant.

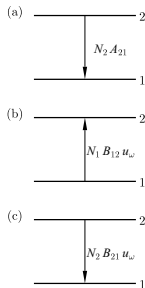


Fig. 23.8 Transitions for a two-level system: (a) spontaneous emission of a photon; (b) absorption of a photon; (c) stimulated emission of a photon.

- Quantum mechanics allows the reverse process to occur. Thus an atom in level 2 can emit a photon of energy $h\omega$ as a direct result of the radiation field, and the atom will end up in level 1 [Fig. 23.8(c)]. In terms of individual photons, this process involves two photons: the presence of a first photon in the radiation field (which is absorbed and then re-emitted) stimulates the emission by the atom of an additional photon. This process is called **stimulated emission**, and will occur at a rate which is proportional both to u_ω and to the number of atoms in level 2. Thus the rate can be written as $N_2 B_{21} u_\omega$ where B_{21} is a constant.

The constants A_{21} , B_{12} and B_{21} are called the **Einstein A and B coefficients**. To summarize, our three processes are:

- (1) spontaneous emission (one photon emitted);
- (2) absorption (one photon absorbed);
- (3) stimulated emission (one photon absorbed, two photons emitted).

In the steady state, with all three processes occurring simultaneously, we must have

$$N_2 B_{21} u_\omega + N_2 A_{21} = N_1 B_{12} u_\omega. \quad (23.58)$$

This can be rearranged to give

$$u_\omega = \frac{A_{21}/B_{21}}{(N_1 B_{12}/N_2 B_{21}) - 1}. \quad (23.59)$$

If the system is in thermal equilibrium, then the relative populations of the two levels must be given by a Boltzmann factor, i.e.

$$\frac{N_2}{N_1} = \frac{g_1}{g_2} e^{-\beta h\omega}, \quad (23.60)$$

where g_1 and g_2 are the degeneracies of levels 1 and 2 respectively. Substitution of eqn 23.60 into eqn 23.59 yields

$$u_\omega = \frac{A_{21}/B_{21}}{(g_1 B_{12}/g_2 B_{21}) e^{-\beta h\omega} - 1}, \quad (23.61)$$

and comparison with eqn 23.42 yields the following relations between the Einstein A and B coefficients:

$$\frac{B_{21}}{B_{12}} = \frac{g_1}{g_2} \quad \text{and} \quad A_{21} = \frac{h\omega^3}{\pi^2 c^3} B_{21}. \quad (23.62)$$

Example 23.4

When will a system of atoms in a radiation field exhibit **gain**, i.e. produce more photons than they absorb?

Solution: The atoms will produce more photons than they absorb if the rate of stimulated emission is greater than the absorption rate, and this will occur if

$$N_2 B_{21} u_\omega > N_1 B_{12} u_\omega, \quad (23.63)$$

which implies that

$$\frac{N_2}{g_2} > \frac{N_1}{g_1}. \quad (23.64)$$

This means that we need to have a **population inversion**, so that the number of atoms ('the population') in the upper state (per degenerate level) exceeds that in the lower state. This is the principle behind the operation of the **laser** (a word that stands for light amplification by stimulated emission of radiation). However, in our two-level system such a population inversion is not possible in thermal equilibrium. For laser operation, it is necessary to have further energy levels to provide additional transitions: these can provide a mechanism to ensure that level 2 is *pumped* (fed by transitions from another level, keeping its population high) and that level 1 can *drain away* (into another lower level, so that level 1 has a low population).

Chapter summary

- The power emitted per unit area of a black body surface at temperature T is given by σT^4 , where

$$\sigma = \frac{\pi^2 k_B^4}{60c^2 h^3} = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}.$$

- Radiation pressure p due to black body photons is equal to $u/3$ where u is the energy density. Radiation pressure due to a collimated beam of light is equal to u .
- The spectral energy density u_ω takes the form of a black body distribution. This form fits well to the experimentally measured form of the cosmic microwave background. It is also important in the theory of lasers.

Further reading

- A discussion of lasers may be found in Foot (2004), chapters 1 and 7.
- More information concerning the cosmic microwave background is in Liddle (2003) chapter 10 and Carroll and Ostlie (1996) chapter 27.

Exercises

- (23.1) The temperature of the Earth's surface is maintained by radiation from the Sun. By making the approximation that the Sun is a black body, but now assuming that the Earth is a grey body with albedo A (this means that it reflects a fraction A of the incident energy), show that the ratio of the Earth's temperature to that of the Sun is given by

$$T_{\text{Earth}} = T_{\text{Sun}}(1 - A)^{1/4} \sqrt{\frac{R_{\text{Sun}}}{2D}}, \quad (23.65)$$

where R_{Sun} is the radius of the Sun and the Earth-Sun separation is D .

- (23.2) Show that the maxima in the functions u_ω and u_λ can be computed by maximising the function $x^\alpha/(e^x - 1)$ for $\alpha = 3$ and $\alpha = 5$ respectively. Show that this implies that

$$x = \alpha(1 - e^{-x}). \quad (23.66)$$

This equation can be solved by iterating

$$x_n = \alpha(1 - e^{-x_{n-1}}); \quad (23.67)$$

now show that (using an initial guess of $x_1 = 1$) this leads to the values given in eqns 23.53 and 23.54.

- (23.3) The cosmic microwave background (CMB) radiation has a temperature of 2.73 K.
- What is the photon energy density in the Universe?
 - Estimate the number of CMB photons which fall on the outstretched palm of your hand every second.
 - What is the average energy due to CMB radiation which lands on your outstretched palm every second?
 - What radiation pressure do you feel from CMB radiation?

- (23.4) What is the ratio of the number of photons from the Sun to the number of CMB photons which irradiate your outstretched hand every second (during the daytime!)?

- (23.5) Thermal radiation can be treated thermodynamically as a gas of photons with internal energy $U = u(T)V$ and pressure $p = u(T)/3$, where $u(T)$ is the energy density. Show that:

- the entropy density s is given by $s = 4p/T$;
- the Gibbs function $G = 0$;
- the heat capacity at constant volume $C_v = 3s$ per unit volume;

- (d) the heat capacity at constant pressure, C_p , is infinite. (What on earth does that mean?)

- (23.6) Ignoring the zero-point energy, show that the partition function Z for a gas of photons in volume V is given by

$$\ln Z = -\frac{V}{\pi^2 c^3} \int_0^\infty \omega^2 \ln(1 - e^{-\hbar\omega/k_B T}) d\omega, \quad (23.68)$$

and hence, by integrating by parts, that

$$\ln Z = \frac{V \pi^2 (k_B T)^3}{45 \hbar^3 c^3}. \quad (23.69)$$

Hence show that

$$F = -\frac{4\sigma VT^4}{3c} \quad (23.70)$$

$$S = \frac{16\sigma VT^3}{3c} \quad (23.71)$$

$$U = \frac{4\sigma VT^4}{3c} \quad (23.72)$$

$$p = \frac{4\sigma T^4}{3c}, \quad (23.73)$$

and hence that $U = -3F$, $pV = U/3$ and $S = 4U/3T$.

- (23.7) Show that the total number N of photons in black body radiation contained in a volume V is

$$N = \int_0^\infty \frac{g(\omega) d\omega}{e^{\hbar\omega/k_B T} - 1} = \frac{2\zeta(3)}{\pi^2} \left(\frac{k_B T}{\hbar c} \right)^3 V, \quad (23.74)$$

where $\zeta(3) = 1.20206$ is a Riemann-zeta function (see Appendix C.4). Hence show that the average energy per photon is

$$\frac{U}{N} = \frac{\pi^4}{30\zeta(3)} k_B T = 2.701 k_B T, \quad (23.75)$$

and that the average entropy per photon is

$$\frac{S}{N} = \frac{2\pi^4}{45\zeta(3)} k_B = 3.602 k_B. \quad (23.76)$$

The result for the internal energy of a photon gas is therefore $U = 2.701 N k_B T$, whereas for a classical ideal gas one obtains $U = \frac{3}{2} N k_B T$. Why should the two results be different? Compare the expression for the entropy of a photon gas with that for an ideal gas (the Sackur-Tetrode equation); what is the *physical* reason for the difference?